Izveštaj o performansama – student 3

Za testiranje performansi korišćen je Locust alat. Zbog nemogućnosti drugačijeg okruženja, testovi su pokrenuti na istom računaru (laptopu) gde je bio pokrenut i backend.

Specifikacije računara:

- CPU: AMD Ryzen 7 5800H with Radeon Graphics, 3201 Mhz, 8 Core(s), 16 Logical Processor(s)
- RAM: 16GB
- GPU: NVIDIA GeForce RTX 3050 Ti Laptop GPU + integrisana AMD

Za testove su izabrani endpointovi koji se najčešće koriste. Nisu izabrani update endpointovi jer bi zbog optimistic locking-a dolazilo do velikog failure rate-a.

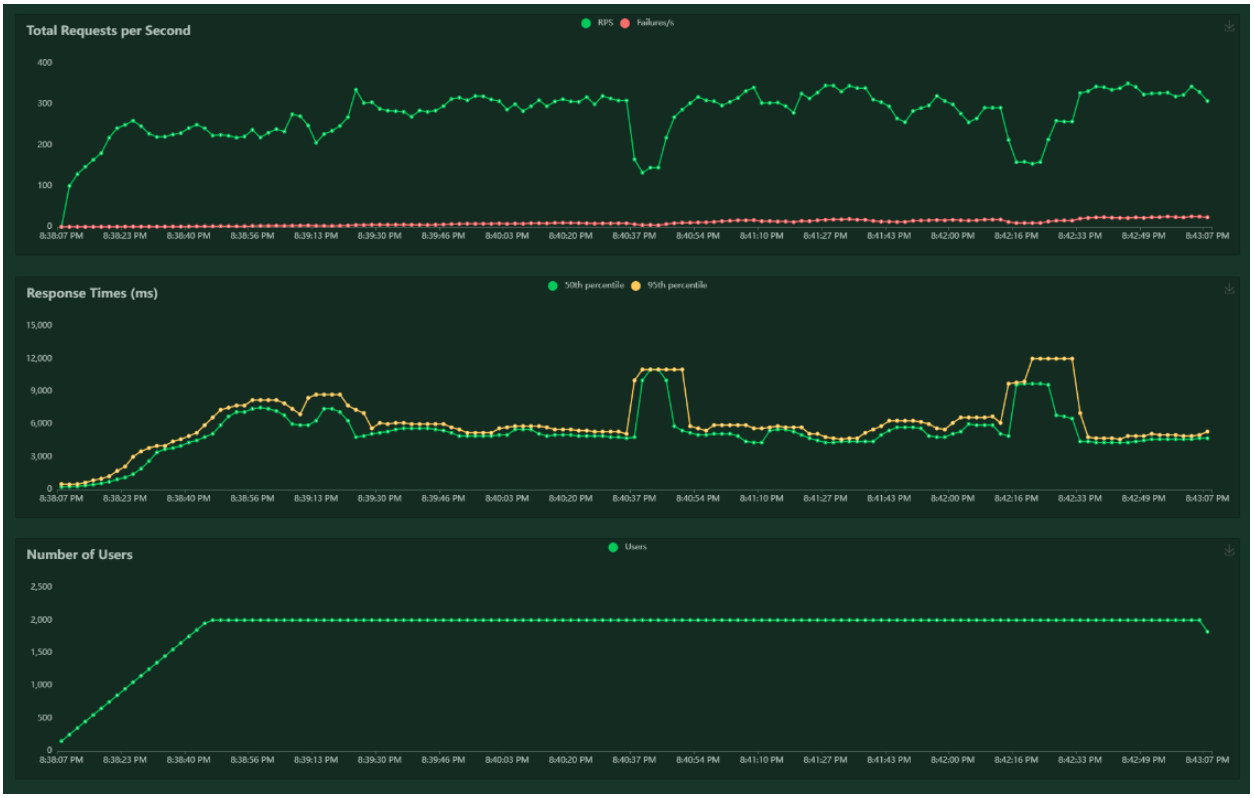
Testovi su rađeni sa 2000 max usera, spawn rate 50, pušteni da rade ~5 minuta, sa tim da je neke sam Locust prekinuo oko 4 minuta zbog prevelikog opterećenja CPU-a.

Za svaki endpoint su postavljene četiri slike – prve dve pokazuju performanse bez keširanja, naredne dve pokazuju performanse sa keširanjem. Najveća pozitivna promena u performansama je viđena kod velikih datasetova sa dosta referenci na druge entitete (foreign key-ova).

Performanse su varijabilne jer je laptop računar intenzivno korišćen zadnje 4 godine.

1. create_vehicle (POST api/vehicles)

Bez keširanja



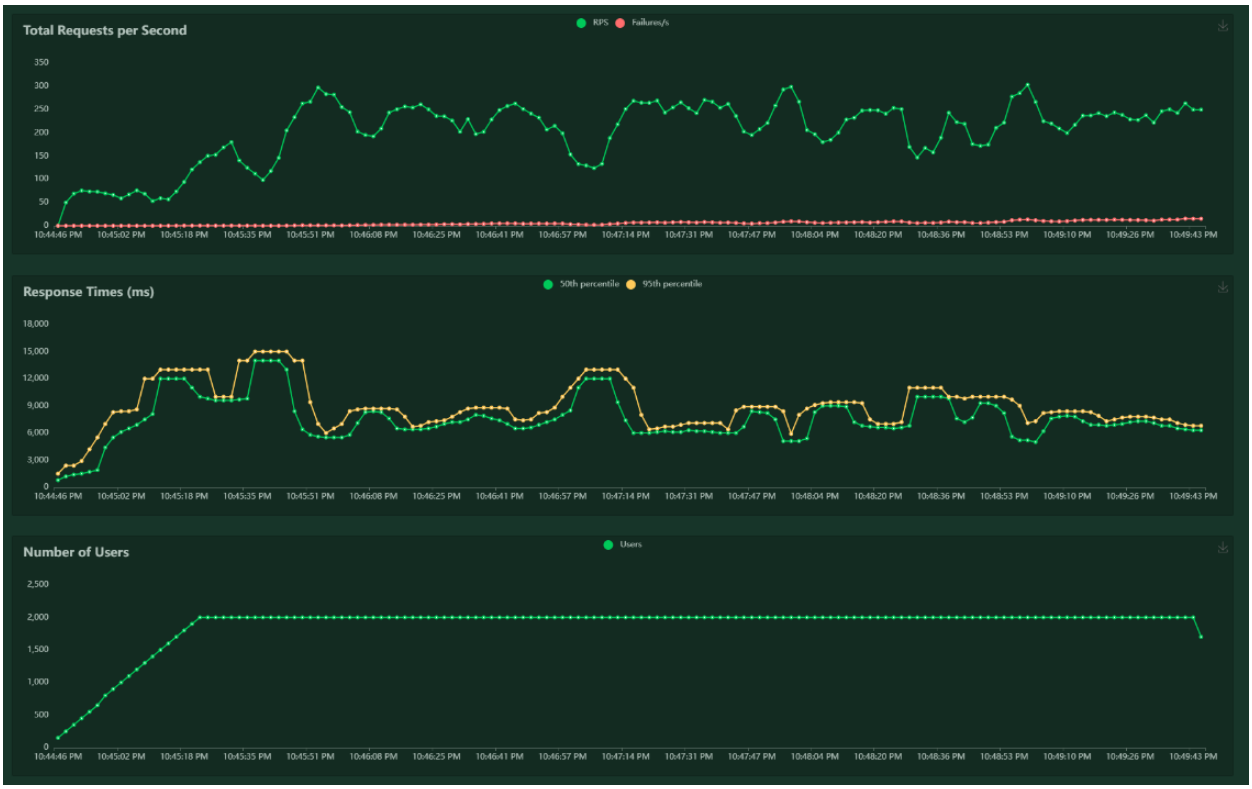
Request Statistics

Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	3057	100	7880	436	6.6	0.0
POST	/api/vehicles	81491	3152	5236	18	12629	390	270.8	10.5
	Aggregated	**83491**	**3152**	**5184**	**18**	**12629**	**391**	**277.5**	**10.5**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	3300	3700	4300	5000	6900	7200	7800	7900
POST	/api/vehicles	5000	5200	5500	5800	7000	8800	11000	13000
	Aggregated	**4900**	**5200**	**5500**	**5800**	**7000**	**8700**	**11000**	**13000**

Sa keširanjem (za ovaj endpoint se ništa ne kešira, nema smis)



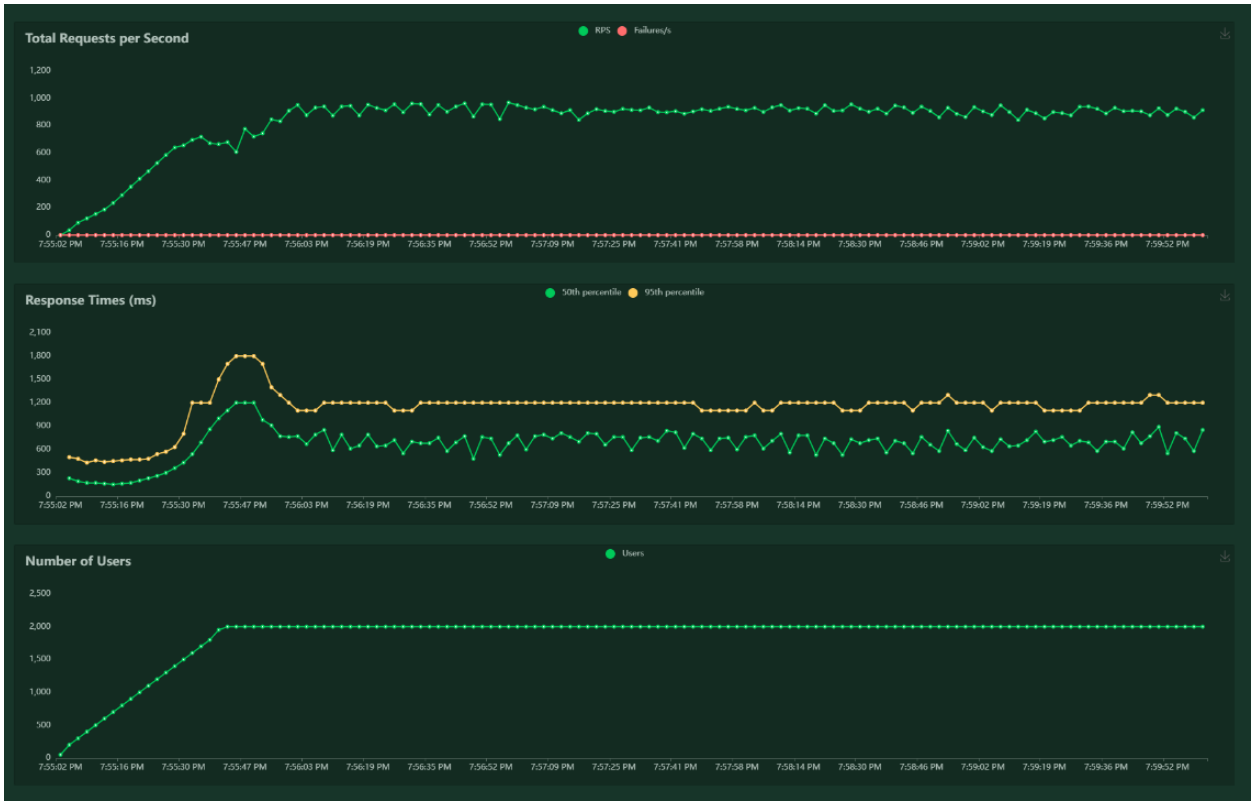
Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	7841	163	14482	436	6.7	0.0
POST	/api/vehicles	60121	1698	7461	362	15864	394	200.2	5.7
	Aggregated	**62121**	**1698**	**7473**	**163**	**15864**	**395**	**206.9**	**5.7**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	9000	9300	10000	11000	12000	13000	14000	14000
POST	/api/vehicles	6900	7300	8000	8700	10000	12000	14000	16000
	Aggregated	**6900**	**7400**	**8100**	**8800**	**10000**	**12000**	**14000**	**16000**

2. get_all_makes (GET api/vehicles/makes)

Bez keširanja



Request Statistics

Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	640	98	3713	436	6.7	0.0
GET	/vehicles/makes	254298	0	637	4	3535	203	847.6	0.0
	Aggregated	**256298**	**0**	**637**	**4**	**3713**	**204**	**854.3**	**0.0**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	440	540	760	1100	1400	1800	2200	3700
GET	/vehicles/makes	690	850	970	1100	1200	1200	1500	3500
	Aggregated	**680**	**850**	**970**	**1100**	**1200**	**1200**	**1500**	**3700**

Sa keširanjem (minimalno povećanje u performansama)

Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	1249	96	6949	436	6.7	0.0
GET	/vehicles/makes	268653	0	523	3	2426	203	894.6	0.0
	Aggregated	**270653**	**0**	**529**	**3**	**6949**	**204**	**901.2**	**0.0**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	450	540	770	1700	4700	5900	6600	6900
GET	/vehicles/makes	520	680	800	900	1000	1100	1200	2400
	Aggregated	**520**	**680**	**800**	**900**	**1000**	**1100**	**1300**	**6900**

3. get_models_for_make (GET api/vehicles/make/{id}/models)

Bez keširanja

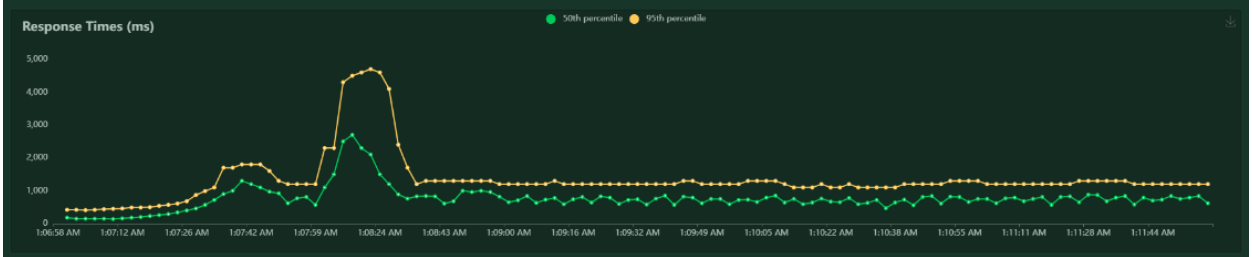
Request Statistics

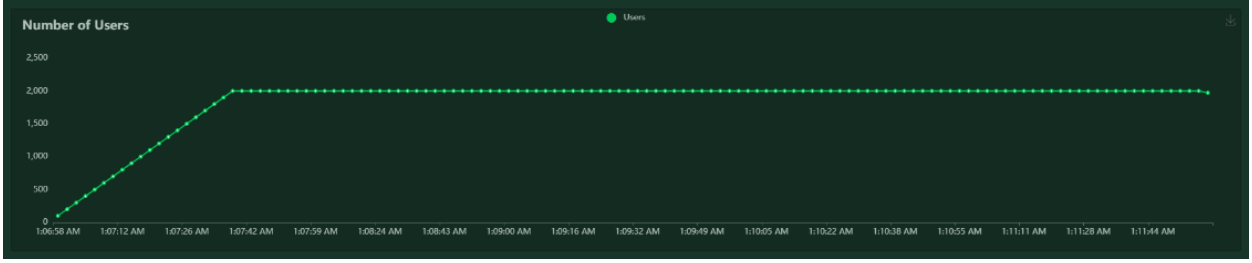
Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	633	94	2614	436	6.7	0.0
GET	/vehicles/make/{makeId}/models	247028	0	694	4	3239	76	821.5	0.0
	Aggregated	**249028**	**0**	**694**	**4**	**3239**	**79**	**828.2**	**0.0**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	440	520	750	1100	1500	1700	2000	2600
GET	/vehicles/make/{makeId}/models	770	930	1100	1100	1200	1300	1500	3200
	Aggregated	**770**	**930**	**1100**	**1100**	**1200**	**1300**	**1600**	**3200**

Sa keširanjem





Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	603	91	2934	436	6.7	0.0
GET	/vehicles/make/{makeId}/models	243408	0	728	4	5326	76	809.5	0.0
	Aggregated	**245408**	**0**	**727**	**4**	**5326**	**79**	**816.2**	**0.0**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	430	500	680	1000	1400	1700	1900	2900
GET	/vehicles/make/{makeId}/models	740	920	1000	1100	1200	1500	3300	5300
	Aggregated	**740**	**920**	**1000**	**1100**	**1200**	**1500**	**3300**	**5300**

4. get_all_vehicles (GET api/vehicles)

Bez keširanja

GET	/api/vehicles?page=6&size=20&sortBy=registrationNumber&direction=asc	13000	14000	15000	15000	16000	16000	16000	16000
GET	/api/vehicles?page=7&size=20&sortBy=registrationNumber&direction=asc	13000	14000	15000	15000	16000	16000	16000	17000
GET	/api/vehicles?page=8&size=20&sortBy=registrationNumber&direction=asc	13000	14000	15000	15000	16000	16000	16000	17000
GET	/api/vehicles?page=9&size=20&sortBy=registrationNumber&direction=asc	13000	14000	15000	15000	16000	16000	16000	16000
	Aggregated	**13000**	**14000**	**15000**	**15000**	**16000**	**16000**	**16000**	**17000**

Sa keširanjem (veliko poboljšanje u performansama)

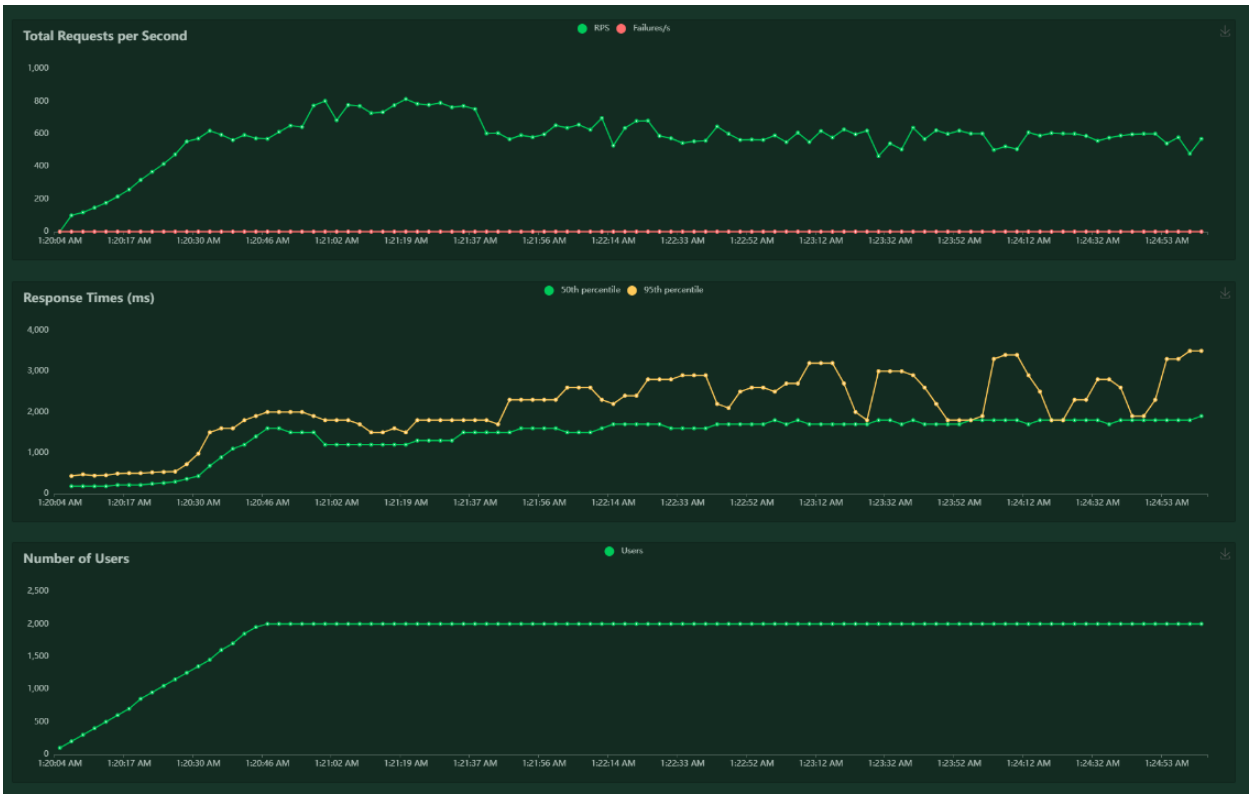
GET	/api/vehicles?page=6&size=20&sortBy=registrationNumber&direction=asc	850	1000	1100	1200	1300	1400	1600	2600
GET	/api/vehicles?page=7&size=20&sortBy=registrationNumber&direction=asc	840	1000	1100	1200	1300	1400	1500	2600
GET	/api/vehicles?page=8&size=20&sortBy=registrationNumber&direction=asc	880	1000	1100	1200	1300	1400	1500	2500
GET	/api/vehicles?page=9&size=20&sortBy=registrationNumber&direction=asc	850	1000	1100	1200	1300	1400	1600	3000
	Aggregated	**860**	**1000**	**1100**	**1200**	**1300**	**1400**	**1500**	**4400**

5. get_vehicle_id (GET api/vehicles/{id})

Bez keširanja

GET	/api/vehicles/10013	1600	1600	2700	2700	2700	2700	2700	2700
GET	/api/vehicles/99997	3600	3600	3600	3600	3600	3600	3600	3600
GET	/api/vehicles/99998	1900	1900	1900	1900	1900	1900	1900	1900
GET	/api/vehicles/99999	1700	1700	1700	1700	1700	1700	1700	1700
	Aggregated	**1900**	**2100**	**2500**	**2900**	**3600**	**3900**	**4300**	**4700**

Sa keširanjem (malo poboljšanje u performansama)



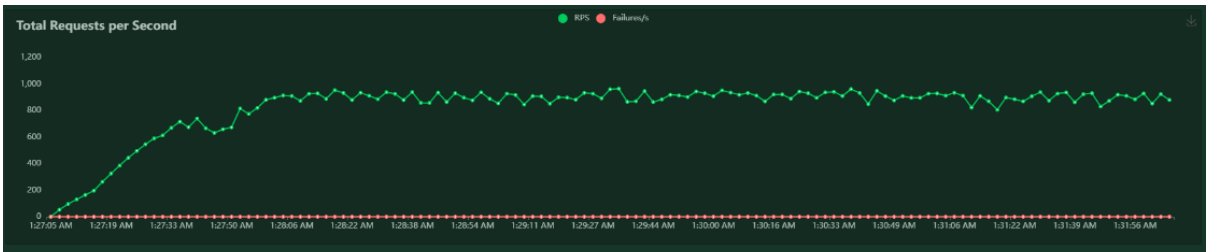
GET	/api/vehicles/1000	1600	1600	1600	1600	1600	1600	1600	1600
GET	/api/vehicles/10000	1400	1400	2200	2200	2200	2200	2200	2200
GET	/api/vehicles/99994	1300	1300	2100	2100	2100	2100	2100	2100
GET	/api/vehicles/99997	1300	1300	1300	1300	1300	1300	1300	1300
GET	/api/vehicles/99998	1700	1700	1700	1700	1700	1700	1700	1700
GET	/api/vehicles/99999	1800	1800	1800	1800	1800	1800	1800	1800
	Aggregated	**1600**	**1700**	**1800**	**1800**	**2200**	**2600**	**3300**	**5000**

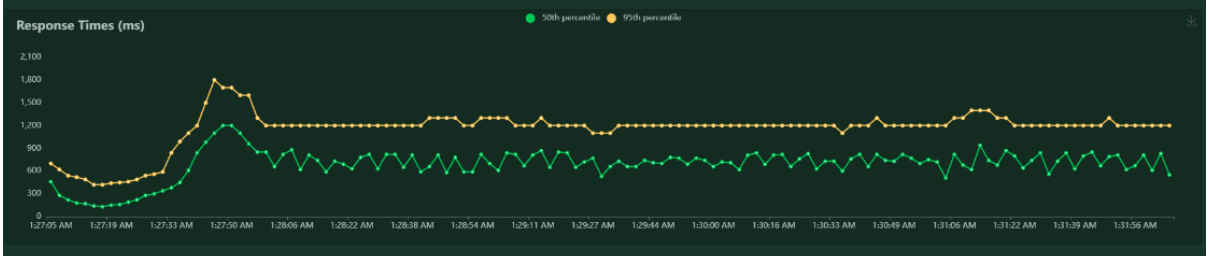
6. search_all_vehicles (GET api/vehicles sa parametrima za filtriranje ili pretragu)

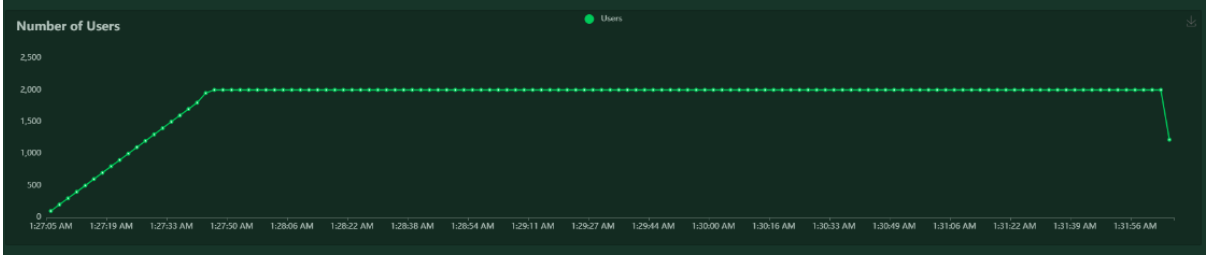
Bez keširanja

GET	/api/vehicles?search=BG&page=0&size=20	40000	41000	41000	41000	42000	42000	43000	44000
GET	/api/vehicles?search=NS&page=0&size=20	39000	40000	40000	41000	41000	42000	42000	43000
	Aggregated	**39000**	**40000**	**40000**	**41000**	**42000**	**42000**	**42000**	**44000**

Sa keširanjem (izuzetno veliko poboljšanje)





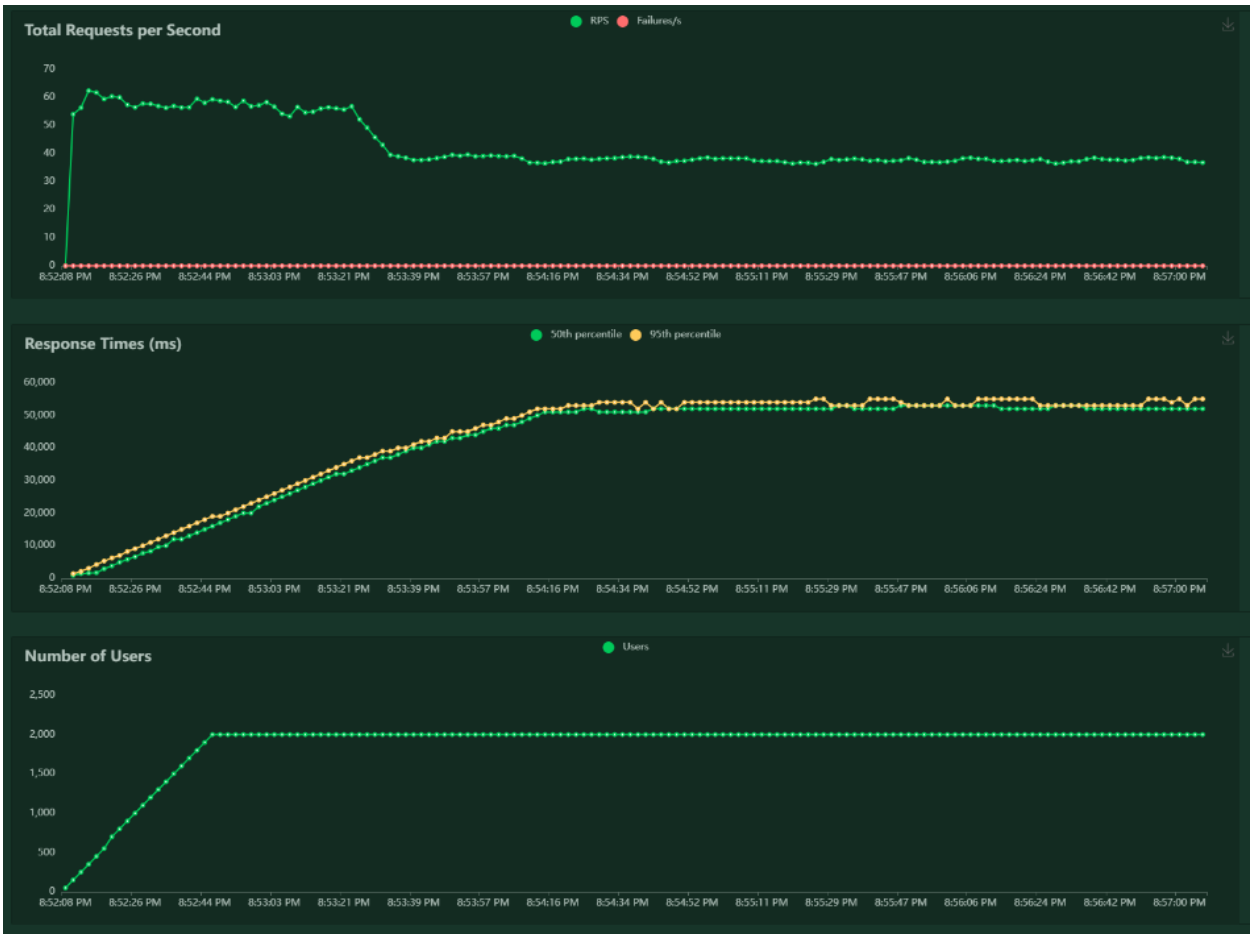


Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	679	97	3869	436	6.7	0.0
GET	/api/vehicles?search=123&page=0&size=20	63554	0	643	4	2449	6744	211.4	0.0
GET	/api/vehicles?search=A&page=0&size=20	63186	0	641	3	2477	1148	210.2	0.0
GET	/api/vehicles?search=BG&page=0&size=20	63255	0	643	4	2476	6677	210.5	0.0
GET	/api/vehicles?search=NS&page=0&size=20	63574	0	638	3	2433	1148	211.5	0.0
	Aggregated	**255569**	**0**	**641**	**3**	**3869**	**3902**	**850.3**	**0.0**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	460	580	820	1200	1500	1900	2200	3900
GET	/api/vehicles?search=123&page=0&size=20	700	880	980	1100	1200	1200	1400	2400
GET	/api/vehicles?search=A&page=0&size=20	700	870	980	1100	1200	1200	1400	2500
GET	/api/vehicles?search=BG&page=0&size=20	700	880	980	1100	1200	1200	1400	2500
GET	/api/vehicles?search=NS&page=0&size=20	680	870	980	1100	1200	1200	1400	2400
	Aggregated	**690**	**880**	**980**	**1100**	**1200**	**1200**	**1500**	**3900**

7. get_my_companies (GET api/companies/my-companies)



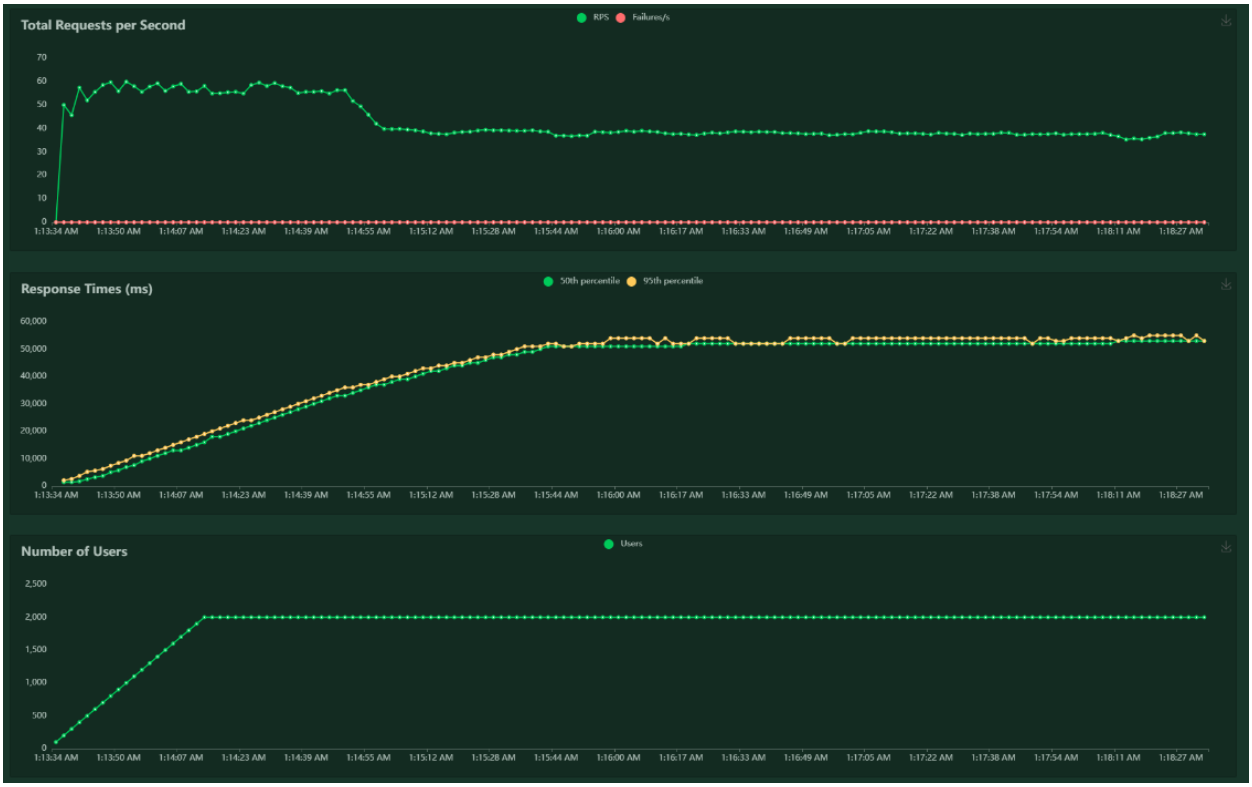
Request Statistics

Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	16108	105	35320	467	6.7	0.0
GET	/companies/my-companies	10796	0	43394	237	60000	2	35.9	0.0
	Aggregated	**12796**	**0**	**39129**	**105**	**60000**	**74**	**42.6**	**0.0**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	16000	19000	23000	27000	30000	32000	34000	35000
GET	/companies/my-companies	52000	52000	52000	52000	53000	53000	55000	60000
	Aggregated	**50000**	**52000**	**52000**	**52000**	**53000**	**53000**	**55000**	**60000**

napisati



Request Statistics

Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	16163	110	34674	467	6.7	0.0
GET	/companies/my-companies	10792	0	43396	315	61230	79	36.0	0.0
	Aggregated	**12792**	**0**	**39138**	**110**	**61230**	**139**	**42.6**	**0.0**

Response Time Statistics

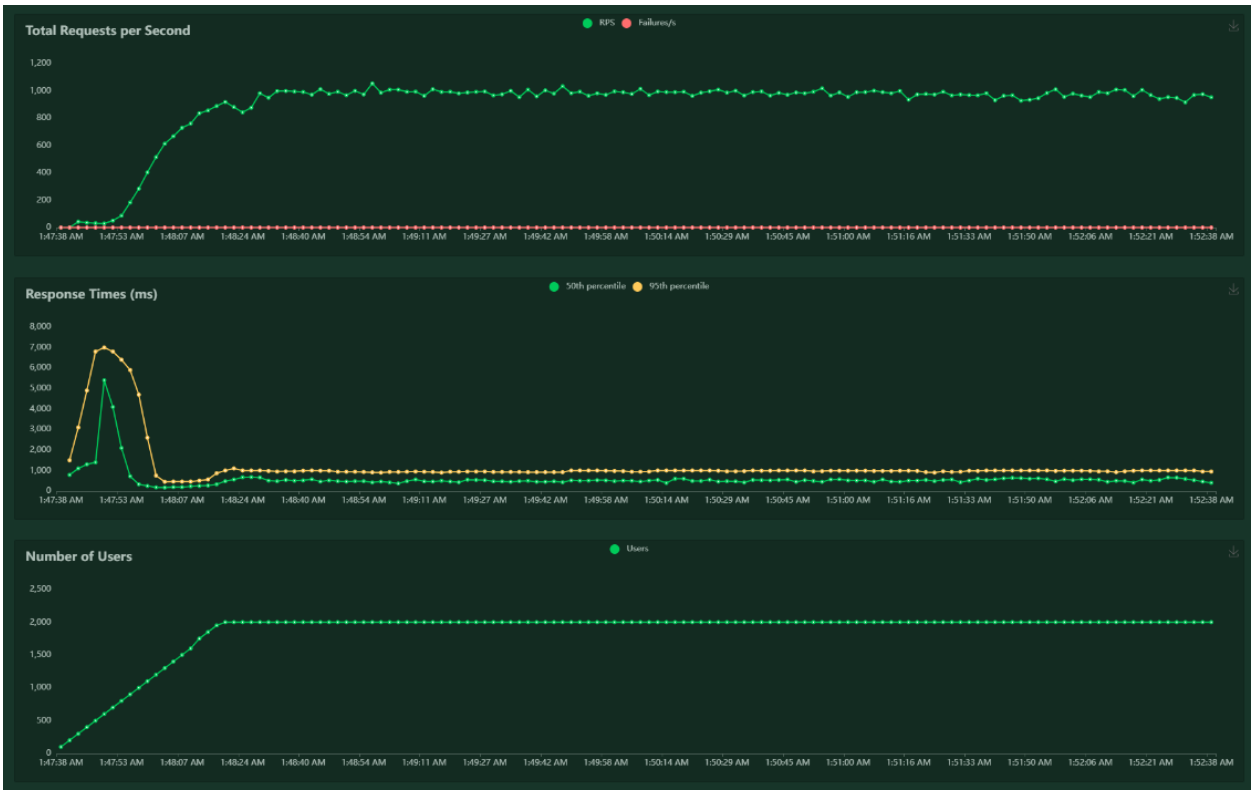
Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	16000	20000	23000	27000	30000	32000	33000	35000
GET	/companies/my-companies	51000	52000	52000	52000	53000	53000	55000	61000
	Aggregated	**50000**	**51000**	**52000**	**52000**	**53000**	**53000**	**55000**	**61000**

8. search_customer_companies (GET api/companies sa parametrima)

Bez keširanja

GET	/api/companies?countryId=8&page=0&size=20	92000	103000	114000	116000	116000	116000	117000	123000
GET	/api/companies?countryId=9&page=0&size=20	90000	102000	113000	116000	116000	117000	121000	122000
GET	/api/companies?countryId=None&page=0&size=20	91000	103000	112000	116000	116000	117000	117000	123000
	Aggregated	**60000**	**81000**	**100000**	**115000**	**116000**	**116000**	**117000**	**127000**

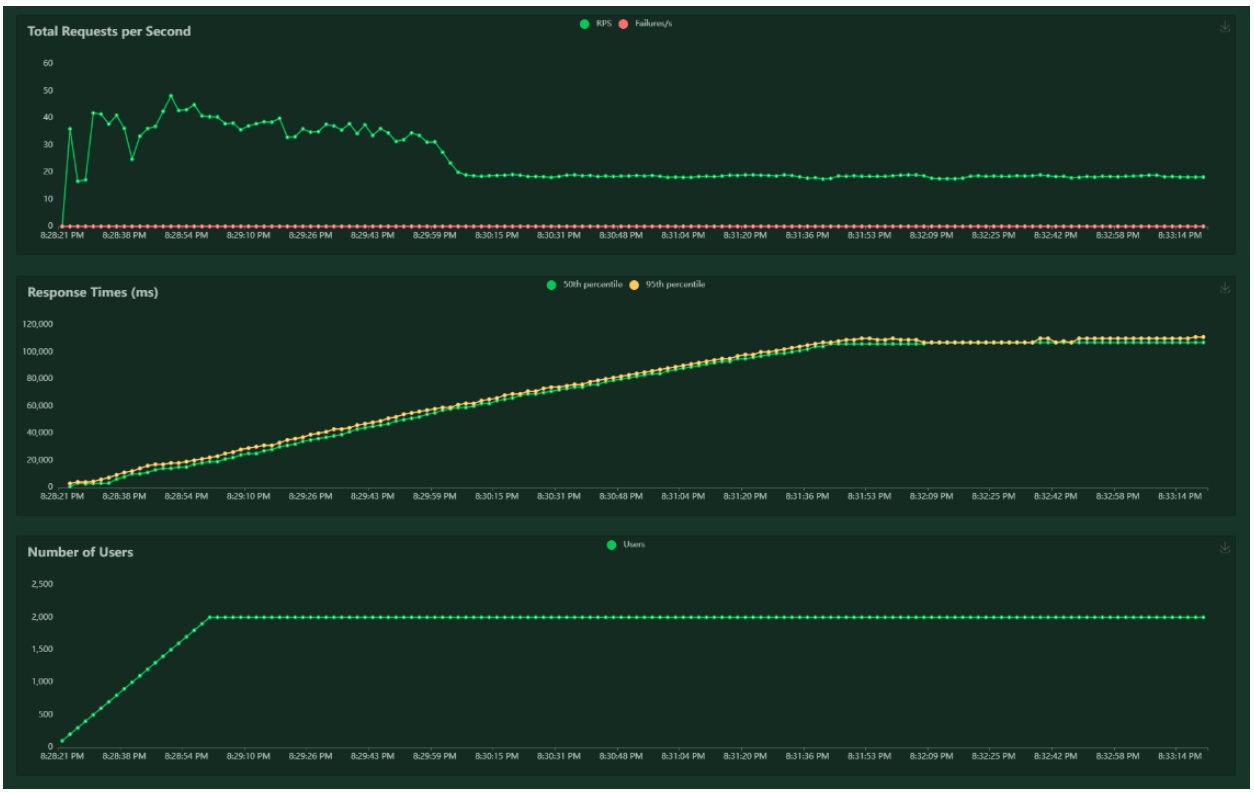
Sa keširanjem (izuzetno veliko poboljšanje)



GET	countryId=6&page=0&size=20	470	620	730	820	930	990	1100	8200
GET	/api/companies?countryId=7&page=0&size=20	470	630	730	820	930	990	1100	8200
GET	/api/companies?countryId=8&page=0&size=20	470	620	730	820	930	990	1100	6500
GET	/api/companies?countryId=9&page=0&size=20	470	620	730	820	930	990	1100	7400
GET	/api/companies?countryId=None&page=0&size=20	470	630	730	820	930	990	1100	6800
	Aggregated	**470**	**630**	**730**	**830**	**940**	**990**	**1200**	**8200**

9. get_pending_companies (GET api/companies/pending)

Bez keširanja



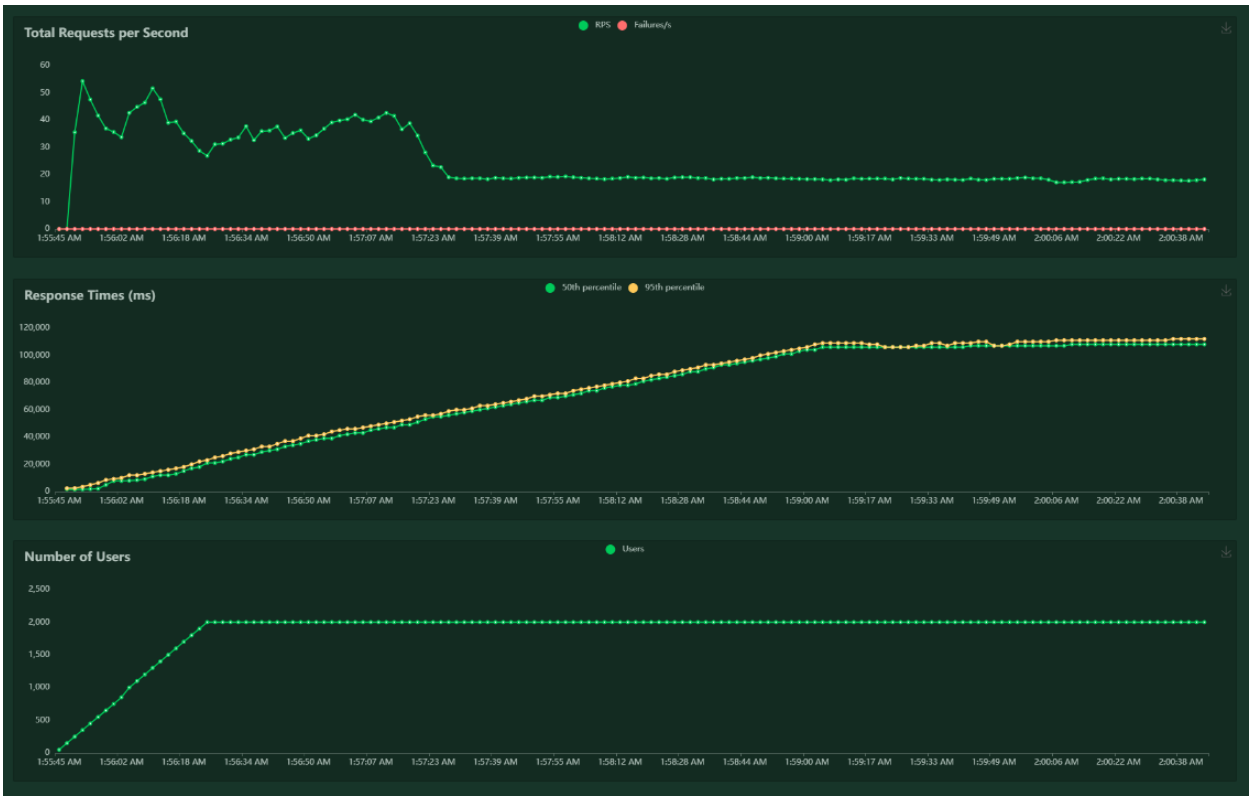
Request Statistics

Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	23810	106	54076	436	6.7	0.0
GET	/companies/pending	5280	0	75658	1877	114145	333	17.6	0.0
	Aggregated	**7280**	**0**	**61414**	**106**	**114145**	**361**	**24.3**	**0.0**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	22000	26000	34000	40000	46000	50000	53000	54000
GET	/companies/pending	87000	99000	106000	107000	107000	107000	110000	114000
	Aggregated	**60000**	**79000**	**98000**	**106000**	**107000**	**107000**	**110000**	**114000**

Sa keširanjem (konkretno ovde se ništa ne kešira, jer u realnoj situaciji nema smisla)



Request Statistics

Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	24753	139	49163	436	6.7	0.0
GET	/companies/pending	5277	0	75074	458	115267	335	17.6	0.0
	Aggregated	**7277**	**0**	**61244**	**139**	**115267**	**363**	**24.2**	**0.0**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	27000	33000	38000	42000	46000	47000	49000	49000
GET	/companies/pending	85000	100000	106000	107000	108000	108000	111000	115000
	Aggregated	**59000**	**77000**	**98000**	**106000**	**107000**	**108000**	**111000**	**115000**

10. get_company_id (GET api/companies/{id})

Bez keširanja

GET	/api/companies/9996	730	730	730	3100	3100	3100	3100	3100
GET	/api/companies/9997	1600	1600	1600	1800	1800	1800	1800	1800
GET	/api/companies/9998	1400	1400	1400	1400	1400	1400	1400	1400
GET	/api/companies/9999	1100	1100	1100	1100	1100	1100	1100	1100
	Aggregated	**1600**	**1700**	**1800**	**1900**	**2300**	**2500**	**3100**	**7500**

Sa keširanjem

Request Statistics

Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
POST	/api/auth/login	2000	0	1865	94	11902	467	6.7	0.0
GET	/api/companies/1	5	0	1157	521	1513	221	0.0	0.0
GET	/api/companies/10	5	0	1324	719	1603	206	0.0	0.0
GET	/api/companies/100	3	0	838	569	1136	207	0.0	0.0
GET	/api/companies/9998	7	0	1207	38	2347	241	0.0	0.0
GET	/api/companies/9999	6	0	1076	613	1591	242	0.0	0.0
	Aggregated	**197659**	**0**	**1181**	**2**	**14386**	**244**	**658.3**	**0.0**

Response Time Statistics

Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
POST	/api/auth/login	620	890	1500	2500	6600	8900	11000	12000
GET	/api/companies/1	1300	1400	1400	1500	1500	1500	1500	1500
GET	/api/companies/10	1400	1500	1500	1600	1600	1600	1600	1600
GET	/api/companies/9998	1300	1300	1300	1400	2300	2300	2300	2300
GET	/api/companies/9999	1200	1200	1200	1200	1600	1600	1600	1600
	Aggregated	**1300**	**1400**	**1400**	**1500**	**1700**	**2000**	**2400**	**14000**